

THE DIGITISATION OF THE EXISTING UGANDA SIGN LANGUAGE, ESTABLISHMENT OF A FORMAL NATIONAL STANDARD FOR UGANDA SIGN LANGUAGE, AND CAPACITY BUILDING TO ENHANCE ACCESS TO AND USE OF DIGITAL SERVICES FOR THE DEAF COMMUNITY IN UGANDA INCEPTION REPORT

Funded by Uganda Communications Commission(UCC) through the Uganda Communications Universal Service and Access Fund (UCUSAF)

Lead Applicant: Uganda National Association of the Deaf (UNAD)

Co-Applicant: Eight Tech Consults Ltd (8TECH)



15%

of the world's population lives with a disability;

80% are in developing countries (WHO)

12.4%

of Ugandans live with some form of disability (UBOS 2016) approximately 4.5million people

4.7%

of Ugandans (2+ yrs) live with hearing difficulty 1,979,613 people (UBOS 2025). With Busoga having the highest prevalence rate of **6.9%**

- ❑ Uganda Sign Language (UGSL) is the primary channel through which Deaf Ugandans access information, education, justice, health and public services.
- ❑ Despite constitutional and legal recognition, UGSL remains insufficiently standardized, inconsistently used across regions, poorly documented, and unsupported by digital tools, limiting Deaf persons' full participation in family, community and enjoyment of human rights entitled to all members of society.
- ❑ This project aims to bridge these gaps by digitalizing UGSL, thereby enhancing accessibility, inclusivity, and communication for deaf individuals.

Main Objective:

Advance inclusive communication and the socio-economic empowerment of Deaf individuals in Uganda by enhancing accessibility and digital literacy, accelerating the adoption of Uganda Sign Language (UGSL) through innovative digital solutions and targeted capacity-building interventions.”



Project Objectives

1. Conduct a National Baseline Study

Assess UGSL access and use, digital literacy levels, and priority support needs within the Deaf community.



2. Standardization

Propose a formal national UGSL standard, including a standardised manual alphabet and linguistic reference descriptions.



3. Application Engineering

Design and develop a UGSL mobile and web application with an interactive dictionary, tutorials and learning modules.



4. Repository Infrastructure

Establish a centralised, publicly accessible UGSL digital repository as the national learning and reference hub.



5. Deploy Assistive Technologies

Integrate digital sign-language tools into public service delivery in education, health and justice.

Project Objectives Cont'd

1

Promote PWD Observatory Uptake

Among Deaf-led organisations,
to strengthen data use for
planning and accountability.

3

Ecosystem Sustainability

Partner with ICT innovators and
embed solutions in national digital
and disability-inclusive policy.

2

Build Individual & Institutional Capacity

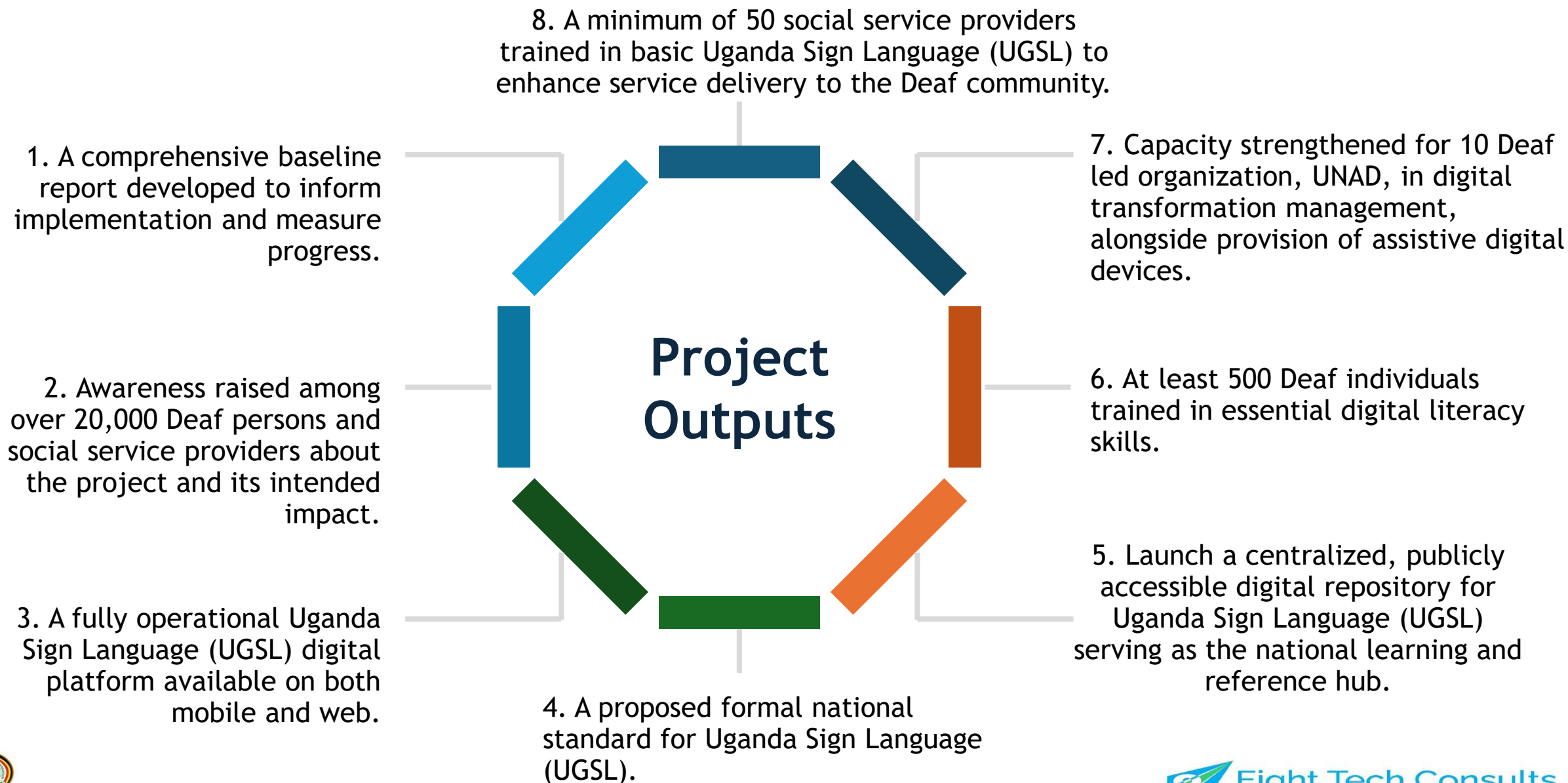
Training-of-Trainers programmes
to use UGSL in accessing, delivering
and managing services.

4

Advocacy & Outreach

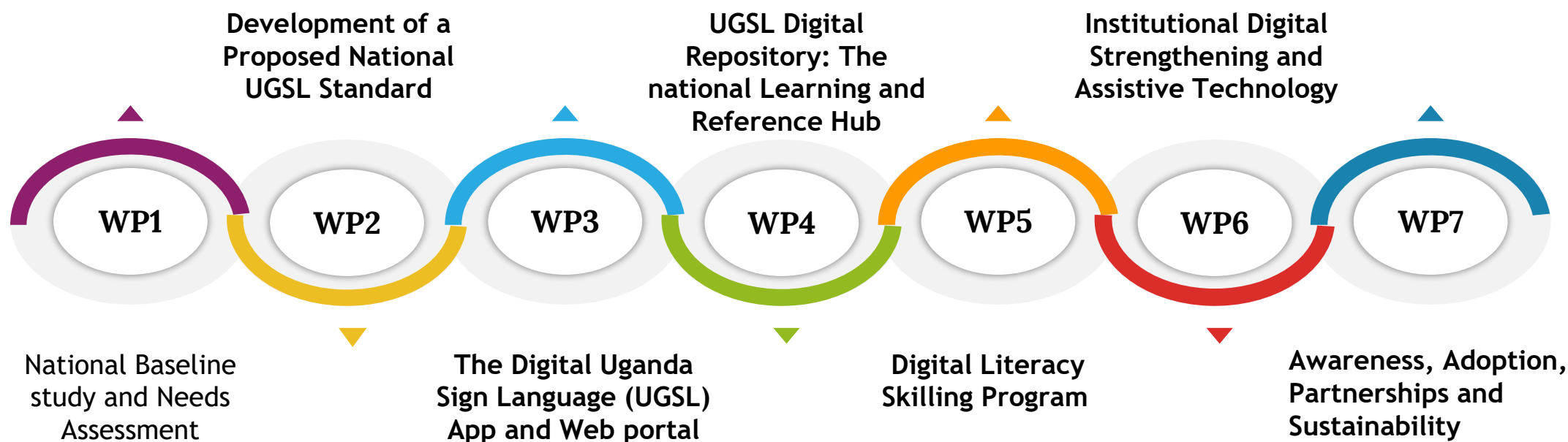
Drive awareness and adoption
of digital UGSL through targeted
campaigns and inter-agency
coordination.

Project Outputs



Approach and Methodology

This project will use a complimentary multidisciplinary knowledge co-creation approach involving seven work packages which include;



Definitions of Key Terms

Term	Definition
Deaf or Deafness:	Refers to individuals with little to no hearing, often using sign language as their primary means of communication.
Uganda Sign Language (UGSL)	The natural visual-gestural language used by the Deaf community in Uganda, recognised under the Constitution of Uganda and the Persons with Disability Act (2020) as the linguistic medium for communication and identity of Deaf people.
Digitalization	The process of converting analogue sign language infrastructure, content and resources into web, mobile and institutional digital formats to enhance accessibility, learning and use of UGSL.
Digital Transformation	The integration of digital technologies into the operations, service delivery and communication systems of the Deaf community and supporting institutions to improve inclusion and efficiency.
Standardization	The establishment of a formal national standard for UGSL, including a standardized manual alphabet and reference descriptions of UGSL phonology, morphology, syntax and semantics.
UGSL Corpus	The existing body of documented Uganda Sign Language vocabulary, signs and reference materials that forms the linguistic basis for digital content and learning materials.
Manual Alphabet	A set of handshapes representing letters of the written alphabet.
UGSL phonology	This refers to the basic "building blocks" of Ugandan Sign Language the smallest visual parts that combine to form a sign. Just as spoken languages use sounds to build words, UGSL uses things like handshape, hand location, hand movement, palm orientation, and facial expression to build signs. Changing any one of these small parts can change the meaning of a sign completely.
Morphology	This is the study of how words (or signs) are formed and how their structure can change their meaning. For example, in UGSL, the way a sign is repeated, sped up, or made larger can change it from a single action to a repeated or intense one.
Syntax	This refers to the rules for how signs or words are arranged to form correct and meaningful sentences.
Semantics	This is the study of meaning what words, signs, or sentences actually mean, and how meaning can change depending on context.
Digital Repository	A centralized, publicly accessible online platform that serves as the national learning and reference hub for all UGSL digital content, e-tutorials and resources.

Definitions of Key Terms Cont'd

Term	Definition
Assistive Technology	Context-appropriate devices, tools and software (such as captioning tools, UGSL-enabled apps and compatible devices) that support Deaf persons in accessing information and services.
Digital Literacy	The essential skills required to use digital devices, applications and online resources effectively, which the project seeks to build among Deaf individuals nationwide.
Training of Trainers (ToT)	A capacity-building model in which selected individuals are trained to become digital literacy champions or instructors who can subsequently train others within their communities.
Persons with Disabilities (PWDs)	Individuals who have long-term physical, mental, intellectual or sensory impairments, including the Deaf, that may hinder their full and effective participation in society.
Deaf-led Organization	An organisation governed and run by Deaf persons, such as UNAD members, that represents and advances the interests of the Deaf community.
Social Service Providers	Personnel working in sectors such as education, health and justice who interact with the Deaf community and are targeted for basic UGSL training to improve service delivery.
PWD Observatory	A national data platform whose uptake among Deaf-led organisations is promoted to strengthen data use for planning and accountability.
Inclusive Communication	Communication that is accessible to and enables the full participation of Deaf individuals on an equal basis with others, removing the linguistic and cultural barriers that exclude them from education, employment, justice, health and public information.
Socio-economic Empowerment	The process of strengthening the social and economic standing of Deaf individuals by improving their access to information, opportunities and services, thereby enabling fuller participation in family, community and national development.
Accessibility	The extent to which information, communication tools, services and digital platforms are designed and made usable by Deaf persons and other persons with disabilities, including features such as captions, animations, visual graphics and UGSL-enabled content.

WP1: National baseline study and Needs Assessment

Aim

Assess UGSL access and use, digital literacy levels, and availability of assistive technologies and inform the requirements for the UGSL app, website and e-education materials.

Approach: Mixed Methods

- Desk review
- Key informant interviews
- Structured surveys
- Focus group discussions

Key Deliverables

- ✓ Approved research design and validated data-collection tools
- ✓ Comprehensive Baseline Report with implementation recommendations
- ✓ Needs-assessment findings feeding into app and website development
- ✓ Dissemination materials and a validation workshop report

WP2: Development of a Proposed National UGSL Standard

A standardised manual alphabet and reference descriptions of UGSL phonology, morphology, syntax and semantics the authoritative linguistic basis for all digital content and learning materials.

Developed through a participatory, knowledge co-creation process led by UNAD's sign language experts, drawing on the existing UGSL corpus, baseline findings, and engagement with the National Curriculum Development Centre, Ministry of Education and Sports, and academic partners.

Development Process

- 1 Scope & Engage**
Convene experts; engage MoGLSD, NCD and Institute of Languages
- 2 Document**
Manual alphabet and linguistic reference descriptions
- 3 Validate**
Iterative expert and community validation
- 4 Consolidate & Submit**
Align with curriculum/policy; submit for formal adoption

WP 3: The Digital UGSL App & Web Portal

Two complementary content models, built using an Agile, user-centred development methodology.



Avatar-Validated Video

Verified UGSL video recordings by Deaf signers serve as the primary, official learning and reference material for the dictionary and tutorials.

Official reference standard



AI-Assisted Dynamic Avatar

Trained on validated UGSL video and glossaries. Piloted with high-frequency content (alphabet, numbers, greetings). Every output undergoes expert review and Deaf-user testing before publication.

Pilot phase — does not replace human-validated content

WP4: UGSL Digital Repository - The National Learning Hub



Centralised & Publicly Accessible

A single national repository serving as the reference hub for UGSL learning content.



Rich Digital Content

E-tutorials, e-books, videos and animations aligned to the national UGSL standard.



Works Offline Too

Content also produced in offline formats DVDs, USBs and memory cards for low-connectivity areas.



Curriculum-Aligned

Developed in partnership with NCDC and the Ministry of Education and Sports.

WP5: Digital Literacy Skilling Programme

500+

Deaf individuals trained

20

Target districts across 5 regions

50+

Social service providers trained

1

Mobilise & Map
Stakeholders

2

Train District
Champions

3

Cascade to Community
Members

4

Train Social Service
Providers

Assistive digital devices and tools are deployed to strengthen 10 Deaf-led organisations, alongside integration of digital sign-language tools directly into public service delivery.

10

Deaf-led organisations
strengthened as digital
“centres of excellence”



Education

Assistive tools
integrated into learning
environments for Deaf
students



Health

UGSL-enabled service
delivery in health
facilities



Justice

Accessible
communication support
within justice service
delivery



Public Launch & Campaigns

Targeted outreach and public campaigns to drive awareness and adoption. Planned in Deaf Awareness Week



PWD Observatory Uptake

Promote uptake among Deaf-led organisations to strengthen data use for planning.



Partnerships & Resource Mobilisation

Explore partnerships with ICT innovators, including NITA-U.



Sustainability & Governance

Embed solutions in national policy; ensure ongoing maintenance and governance.

20,000+

Deaf persons and social
service providers reached
through awareness activities

Workplan Timeline - Quarterly Overview

	Q1 Jul - Sep	Q2 Oct - Dec	Q3 Jan - Mar	Q4 Apr - Jun
Inception & Start-up				
WP1: Baseline Study				
WP2: UGSL Standard				
WP3: App & Web Platform				
WP4: Digital Repository				
WP5: Digital Literacy Skilling				
WP6: Institutional Strengthening				
WP7: Awareness & Sustainability				



The following evaluation stages will be implemented for the Digitalizing Uganda Sign Language (UGSL) project:

- i. Baseline Assessment
- ii. Mid-Term Review
- iii. Endline Evaluation
- iv. Conduct Monitory assessment Post Implementation
- v. A feedback system will be integrated into the platform to allow users to provide input on the usability and effectiveness of the digital content.
- vi. Monthly progress reports will be shared with the UCC Project Implementation Team (PIT) to ensure transparency and accountability.
- vii. Promote a learning culture within the project team and among participants. Use M&E findings to adapt strategies and activities for maximum effectiveness and sustainability as well as provide training and support to project staff.

Risk Management Framework

Key Risk Management Actions:

Proactive Identification and Assessment:

Regular project reviews and stakeholder feedback sessions will be used to identify and assess emerging risks.



Monitoring and Reporting:

Continuous monitoring and quarterly reporting will help to assess the risk status, ensuring that the project adapts to new challenges effectively



Implementation of Mitigation Strategies:

Detailed action plans will be put in place, with timelines, responsibilities, and monitoring mechanisms.



Contingency Planning:

For high-risk areas, alternative strategies will be developed to minimize the impact of unforeseen challenges, including financial resources or additional partnerships.



